

Answer: $\frac{p^2}{m}$

$$i, i+j, \dots, i+mj \pmod{p}$$

Lemma 1. *Let p be a prime and $n < p$ a positive integer. Let $N \subset \mathbb{F}_p$ be a set with $|N| = (p-1)/2$. Let f be the number of pairs $(i, j) \in \mathbb{F}_p^2$ such that*

$$i, i+j, \dots, i+(n-1)j \in N.$$

$$f \leq p^2 \frac{p-1}{n(p+1)+p-1}.$$

Proof. Let $\delta := \frac{p-1}{2p}$, so $|N| = p\delta$. Let

$$A_{ij} := \{i, i+j, \dots, i+(n-1)j\} \text{ and } l_{ij} := |A_{ij} \cap N|,$$

for $i, j \in \mathbb{F}_p$.

Note that for each $k \in \mathbb{F}_p$ and every $j \in \mathbb{F}_p$, there exist exactly n values $i \in \mathbb{F}_p$ such that $k \in A_{ij}$. Thus for each $k \in \mathbb{F}_p$ there exist exactly np pairs $(i, j) \in \mathbb{F}_p^2$ such that $k \in A_{ij}$, so

$$\sum_{i,j} l_{ij} = np|N|.$$

Because $\mathbb{F}_p$ is a field, for each $k_1, k_2 \in \mathbb{F}_p$ with $k_1 \neq k_2$, and every $m_1 \neq m_2$ with $0 \leq m_1, m_2 < n$, there exists a unique pair $(i, j) \in \mathbb{F}_p^2$ such that k_r is the m_r -th element of A_{ij} for $r \in \{1, 2\}$. Thus for each $k_1, k_2 \in \mathbb{F}_p$ with $k_1 \neq k_2$, there exist exactly $\binom{n}{2}$ pairs $(i, j) \in \mathbb{F}_p^2$ such that $k_1, k_2 \in A_{ij}$, so

$$\sum_{i,j} \binom{l_{ij}}{2} = \binom{n}{2} |N|^2.$$

Using $\binom{k}{2} = \frac{k^2-k}{2}$, we find the sum of the squares:

$$\sum_{i,j} l_{ij}^2 = 2 \cdot \binom{n}{2} |N|^2 + np|N| = n(n-1)p^2\delta^2 + np^2\delta.$$

We separate the f sequences for which $l_{ij} = n$ from the remaining $p^2 - f$:

$$\sum_{l_{ij} < n} l_{ij} = np^2\delta - nf \quad \text{and} \quad \sum_{l_{ij} < n} l_{ij}^2 = n(n-1)p^2\delta^2 + np^2\delta - n^2f \quad (1)$$

By the Cauchy-Schwarz inequality, we obtain

$$(p^2 - f) \sum_{l_{ij} < n} l_{ij}^2 \geq \left(\sum_{l_{ij} < n} l_{ij} \right)^2. \quad (2)$$

Plugging equations (1) into (2) and simplifying, we obtain

$$p^2\delta \geq f[n(1-\delta) + \delta].$$

Substituting $\delta = \frac{p-1}{2p}$ gives

$$f \leq p^2 \frac{p-1}{n(p+1)+p-1},$$

as required. $\square$

Given that

$$p^2 \frac{p-1}{(m+1)(p+1)+p-1} < \frac{p^2}{m},$$

from Lemma 1 we immediately obtain that $u(p, m) = \frac{p^2}{m}$ is a valid upper bound. It remains to show that it is optimal up to a multiplicative constant. We use notation defined in the proof of Lemma 1.

Take $N = \{0, 1, 2, \dots, \frac{p-1}{2} - 1\}$. Note that when $a, b \in N$ satisfy $a \leq b$ and $b - a = (n-1)t$ with $t \geq 0$ an integer. Setting $i = a$ and $j = t$, the progression

$$A_{ij} = \{a, a+t, \dots, a+(n-1)t\} = \{a, a+t, \dots, b\}$$

consists of integers lying in the interval $[a, b] \subseteq N$, so $A_{ij} \subseteq N$. Distinct pairs (a, b) give distinct pairs $(i, j) = (a, t)$, since $a = i$ and $b = i + (n-1)j$ are recovered from (i, j) . Thus the number of valid pairs is at least the number of ordered pairs $(a, b) \in N^2$ with $a \leq b$ and $b \equiv a \pmod{n-1}$.

To count these, split N into its residue classes modulo $n-1$. If these classes have sizes $x_1, \dots, x_{n-1}$, then each pair $a \leq b$ within a single class contributes, so by the Cauchy-Schwarz inequality

$$f \geq \sum_{t=1}^{n-1} \binom{x_t}{2} + |N| \geq \sum_{t=1}^{n-1} \frac{x_t^2}{2} \geq \frac{1}{2(n-1)} \left(\sum_{t=1}^{n-1} x_t \right)^2 = \frac{|N|^2}{2(n-1)} = \frac{(p-1)^2}{8(n-1)}.$$

Finally we compare this with the upper bound. Using the trivial estimate $p-1 \geq \frac{p}{2}$ gives

$$f \geq \frac{(p-1)^2}{8(n-1)} \geq \frac{p^2}{32(n-1)} = \frac{p^2}{32m},$$

so the upper bound $u(p, m) = \frac{p^2}{m}$ is optimal up to a multiplicative constant.